# Lecture 14

## Carbon Accounting and Offset Markets

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ECON 467

# Roadmap

1. What are offsets?
2. What issues might they have?
3. How do they work in practice?

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These are called **carbon offsets**

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This can reduce costs since it doesn't really matter *where* we reduce carbon emissions

# Offset markets around the world

Offset markets are currently small but are expected to grow as countries march up their MAC curves

## Global carbon offsets market could be worth $200 billion by 2050: Berenberg

HIGHLIGHTS

**Rapid growth in offsets market set to continue**

**Market no longer to be driven by aviation after virus setback**

**Oil and gas planning to invest heavily over next 5 years: Berenberg**

The value of the global market for carbon emissions offsets could increase to $200 billion by 2050, German bank Berenberg said in a note Wednesday.

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There are (at least) two big sets of carbon offsets:

- Forest carbon offsets
- Clean development mechanism

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The avoided emissions from burned trees (1) or absorbed carbon (2) are the **carbon offsets**

# Forest carbon offsets

ASK MIT CLIMATE

How many new trees would we need to offset our carbon emissions?

How much carbon can a tree absorb?

# Forest carbon offsets

*[...] studies have estimated an average American's carbon footprint at around 16 tons of CO2 annually, one of the highest figures for any country because of the energy-intensive American lifestyle. A single mature tree, meanwhile, may take in about 50 pounds of carbon dioxide per year. At this rate, it would take 640 trees per person to account for all American emissions, which adds up to more than 200 billion trees.*

50 pounds per trees $\times$ 3 trillion trees $=$ 70 billion tons of carbon stored in trees

This is about 2 years worth of emissions

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**Leakage:** preserving or afforesting one area leads to increased deforestation elsewhere

These (especially leakage) are challenges that arise with most environmental policies

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The farmer's optimal decision is given by the usual $PMB(a) = PMC(a)$

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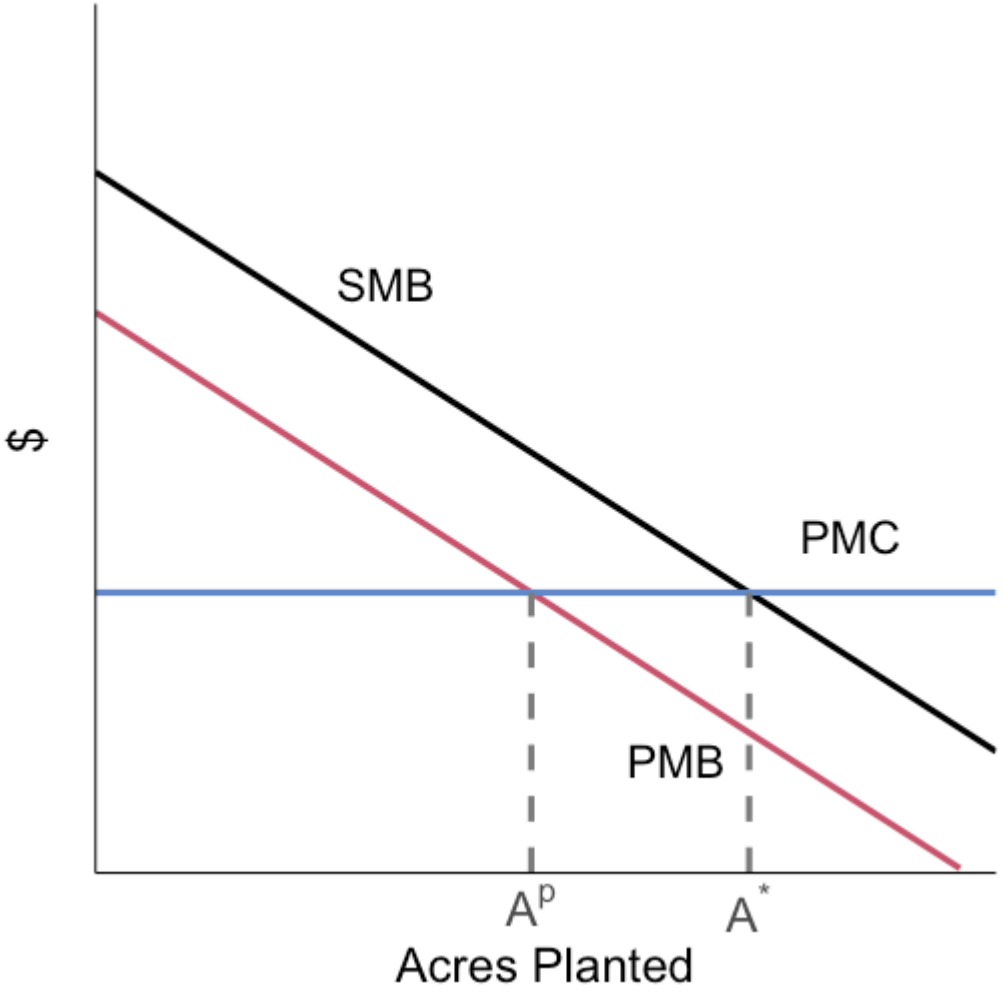


The farmer plants $A^p$

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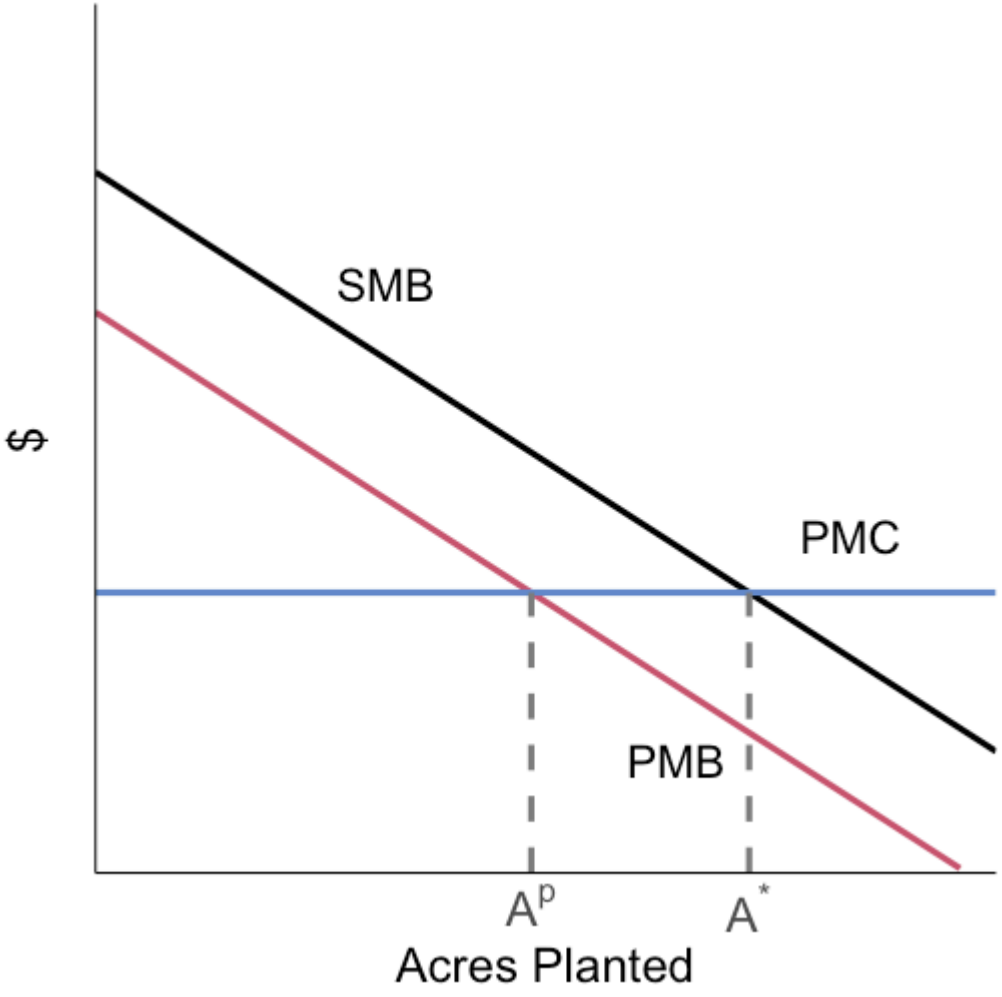


The farmer plants $A^p$

Now, suppose Microsoft offers to pay the farmer some amount of money to plant $A^o$ acres to avoid $A^o$ extra units of carbon

To keep things simple suppose this payment is exactly the marginal external benefit $MEB(a)$

What happens?

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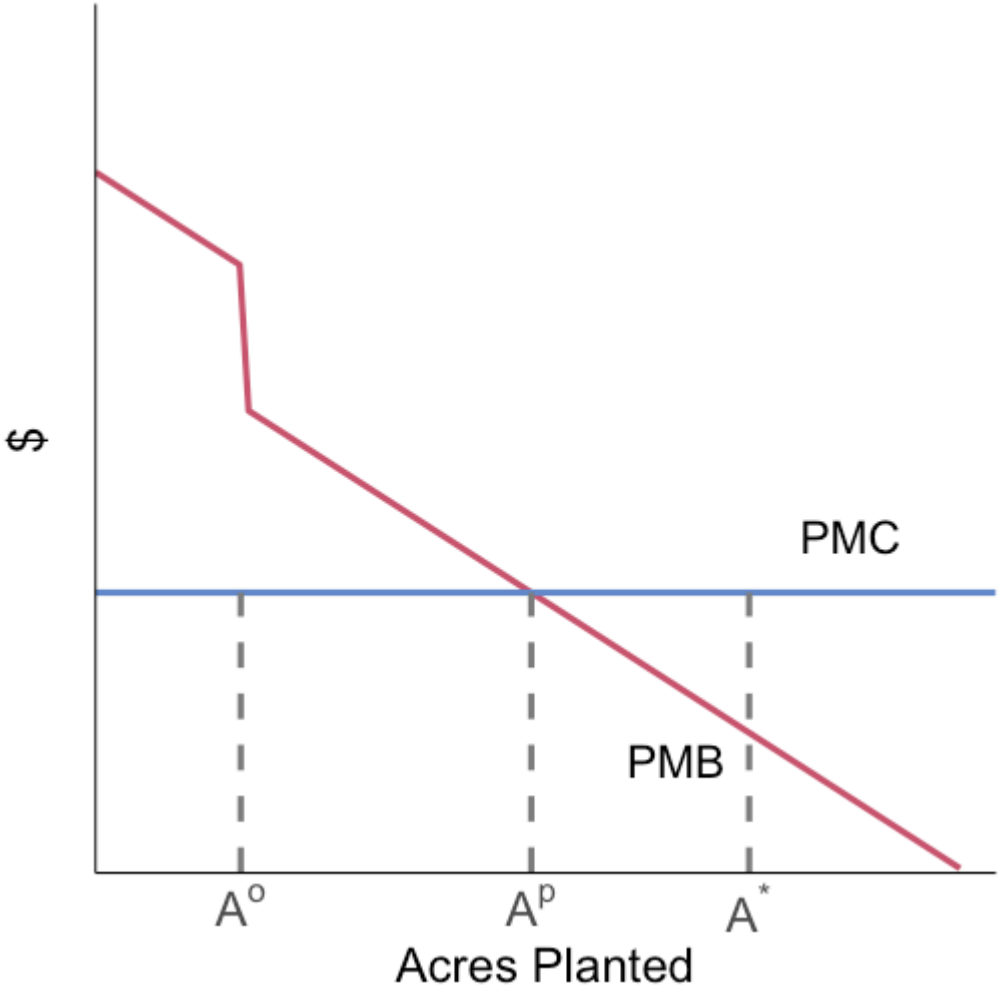


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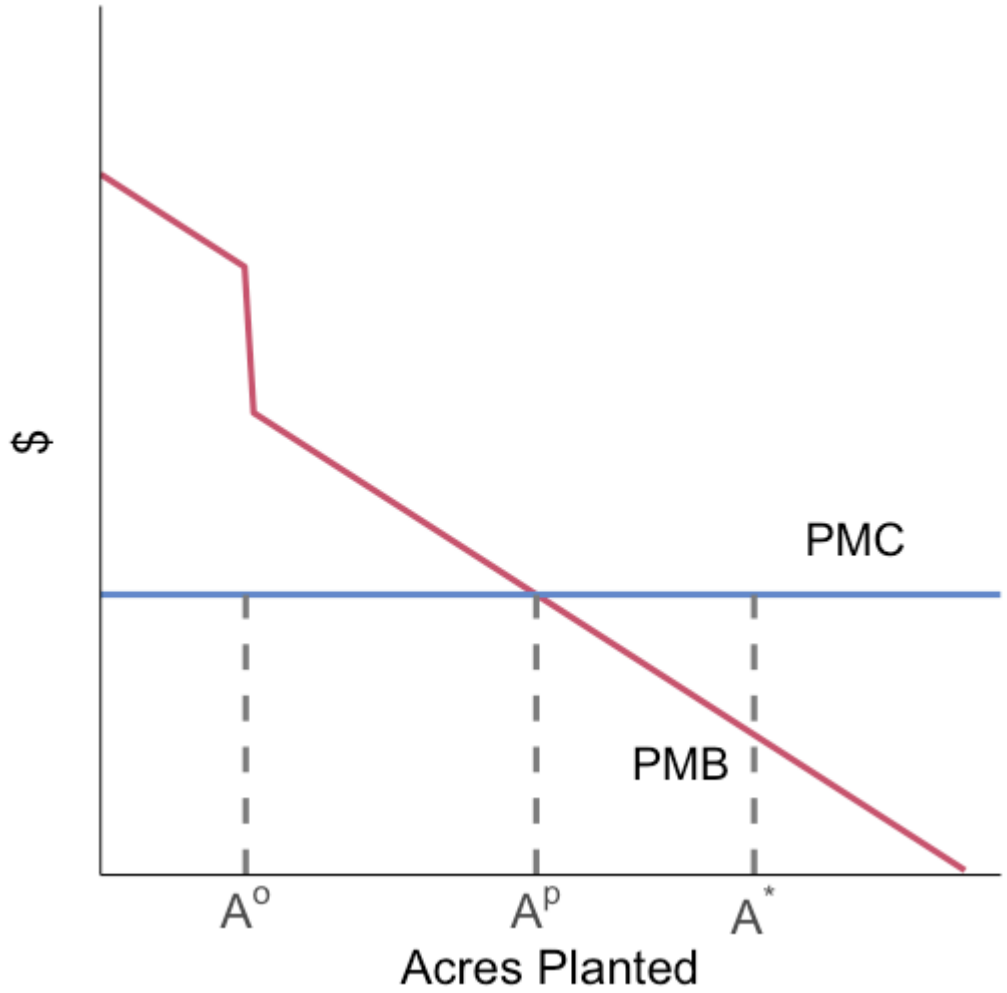


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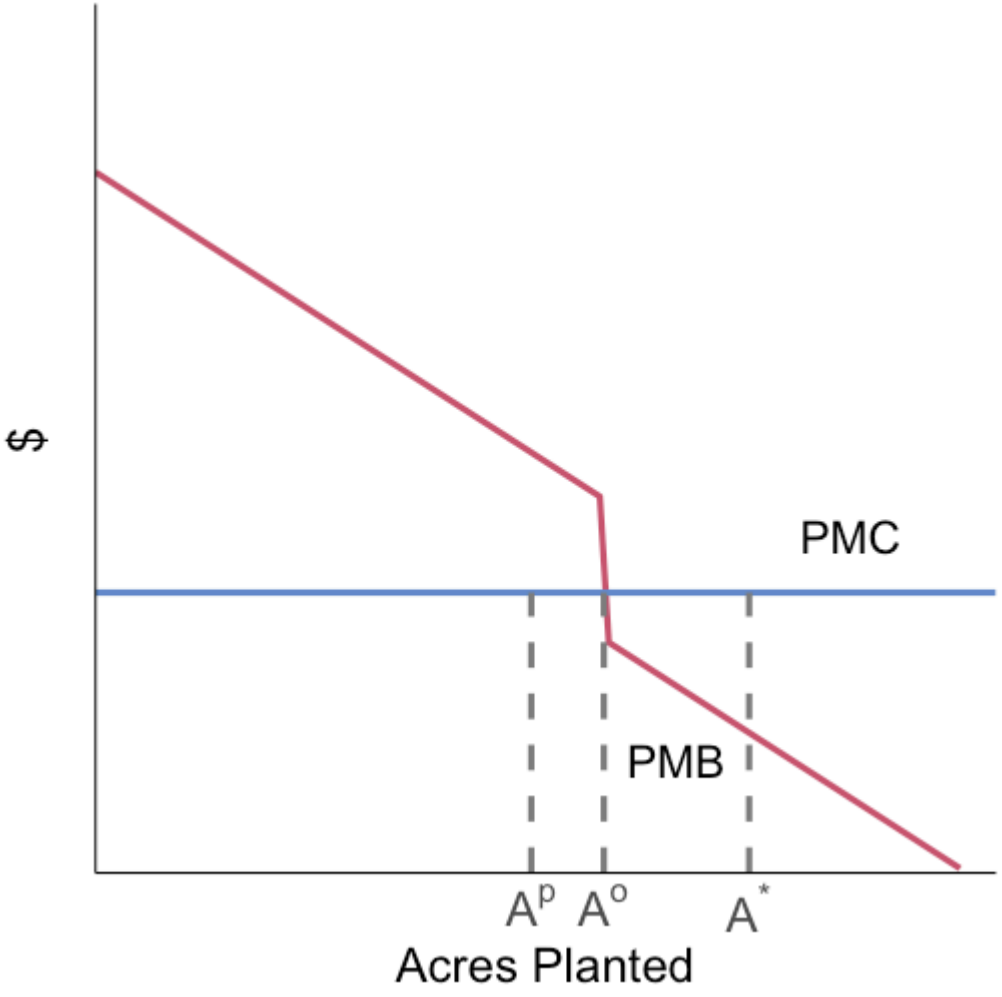
The farmer gets the additional payment for the $A^0$ acres, but the offset did not change the value of the **marginal acre**

The offset payments did not actually increase forest planting at all: was not **additional**

# Challenges with offsets: additionality

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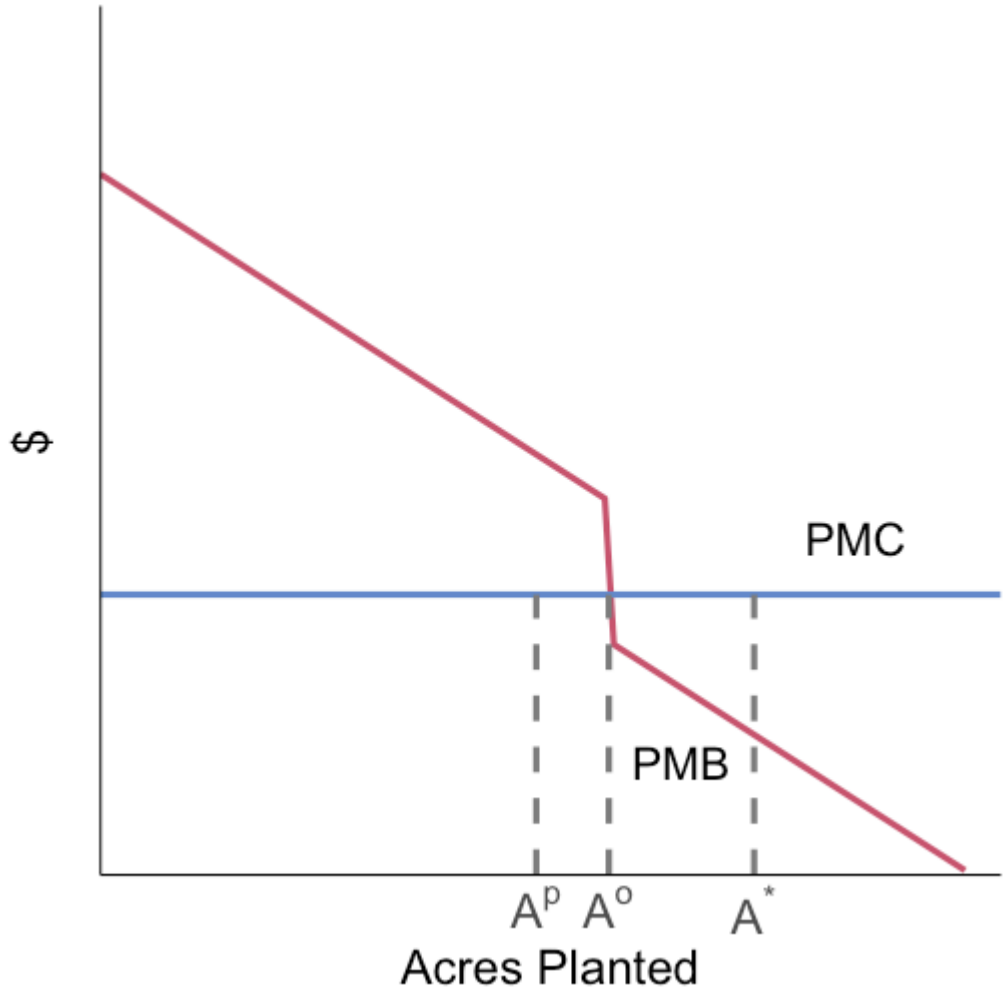


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Paying the farmer for more acres than they otherwise would have planted actually increases forest cover!

**Problem:** we paid the farmer for $A^p$ acres they would have planted anyway and got zero **additional** forest for it

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Incentives need to matter **on the margin** to affect behavior

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The price $p$ clears the cotton market

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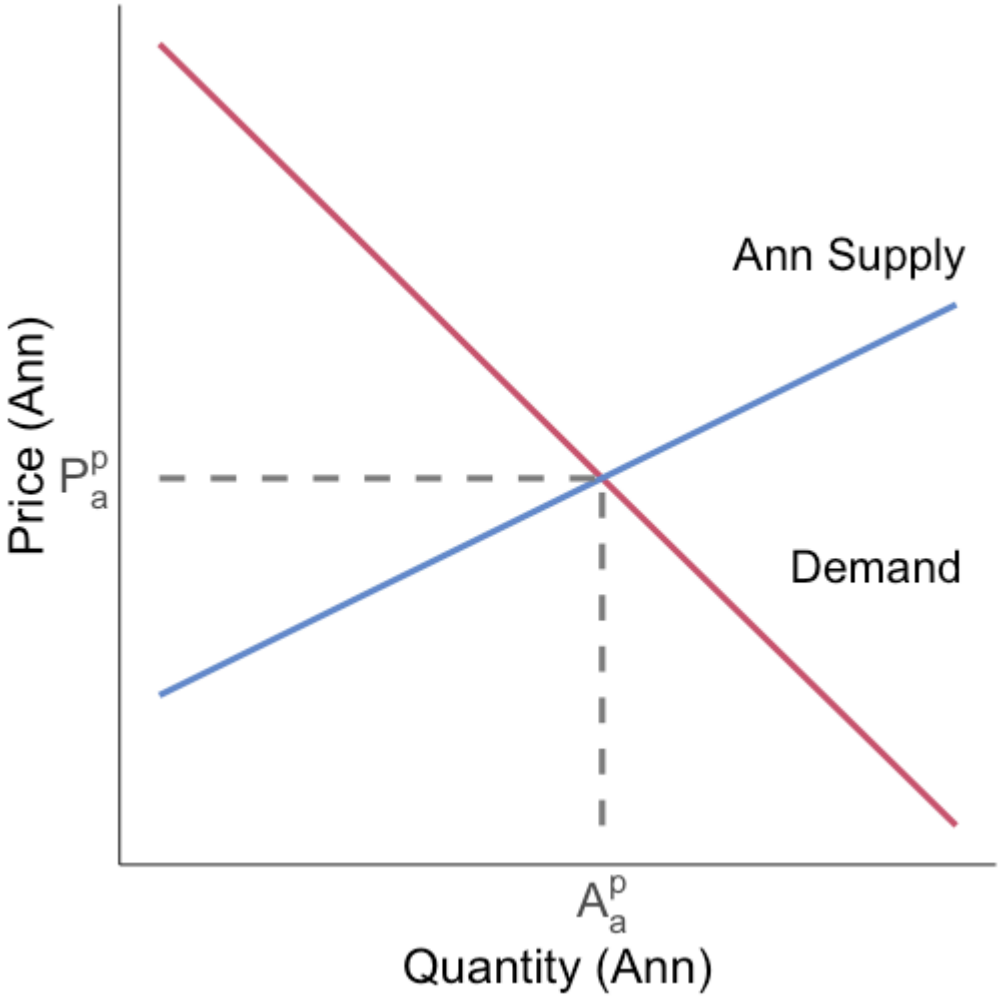


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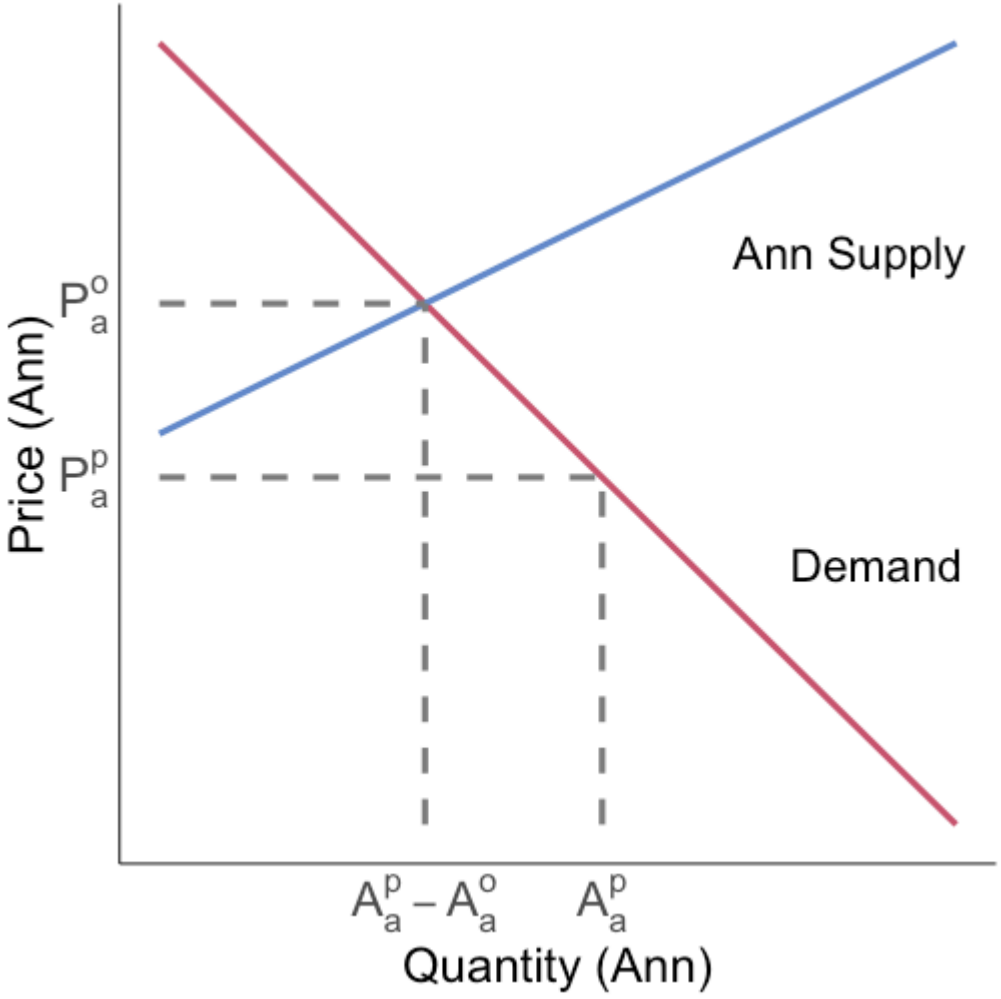



Suppose we pay Ann to keep $A^o$ acres of land as forest

This reduces Ann's supply of cotton by $A^o$, raising the price of cotton to $P^o > P^p$

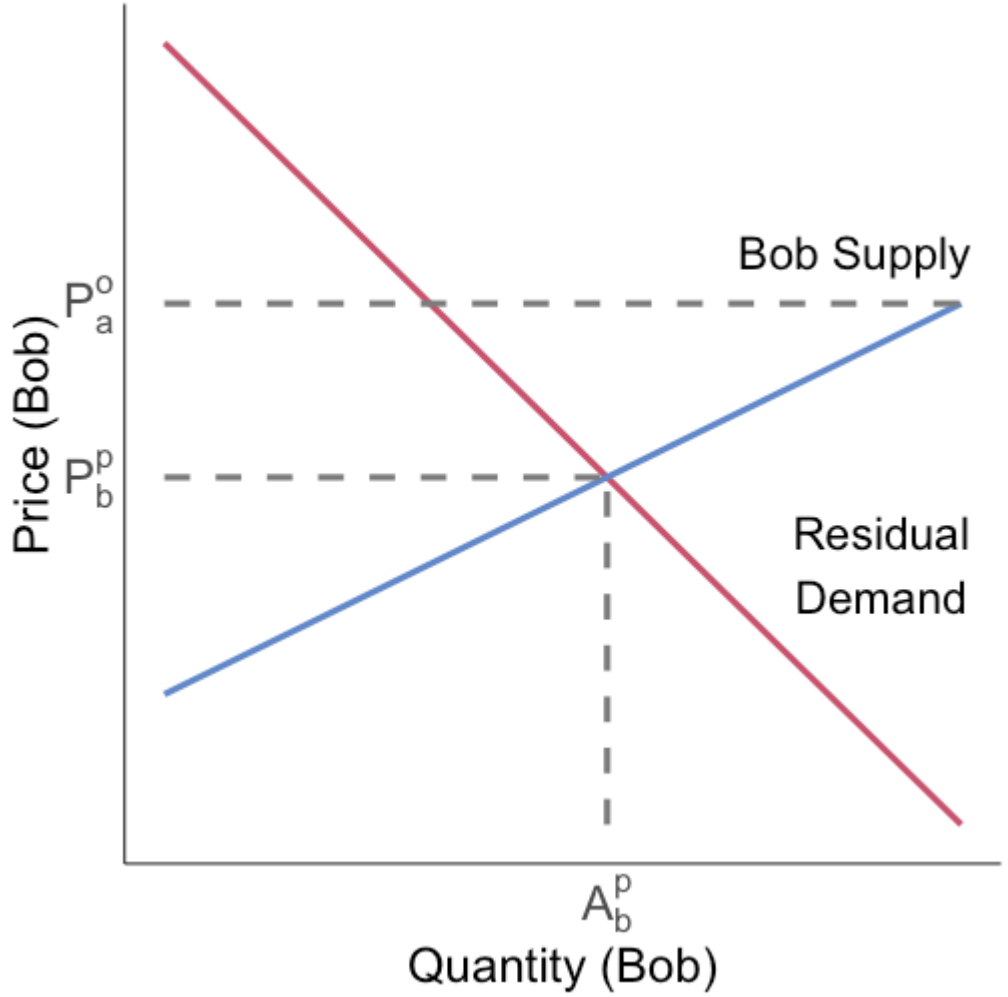
People who were paying more than $P^p$ but less than $P^o$ are no longer buying cotton from Ann

# Challenges with offsets: leakage

Bob has the exact same cost (supply) as Ann

He faces a **residual demand curve:** demand everyone in the market who did not buy from Ann

What happens to Bob when Ann takes $A^o$ acres of land out of cotton production?



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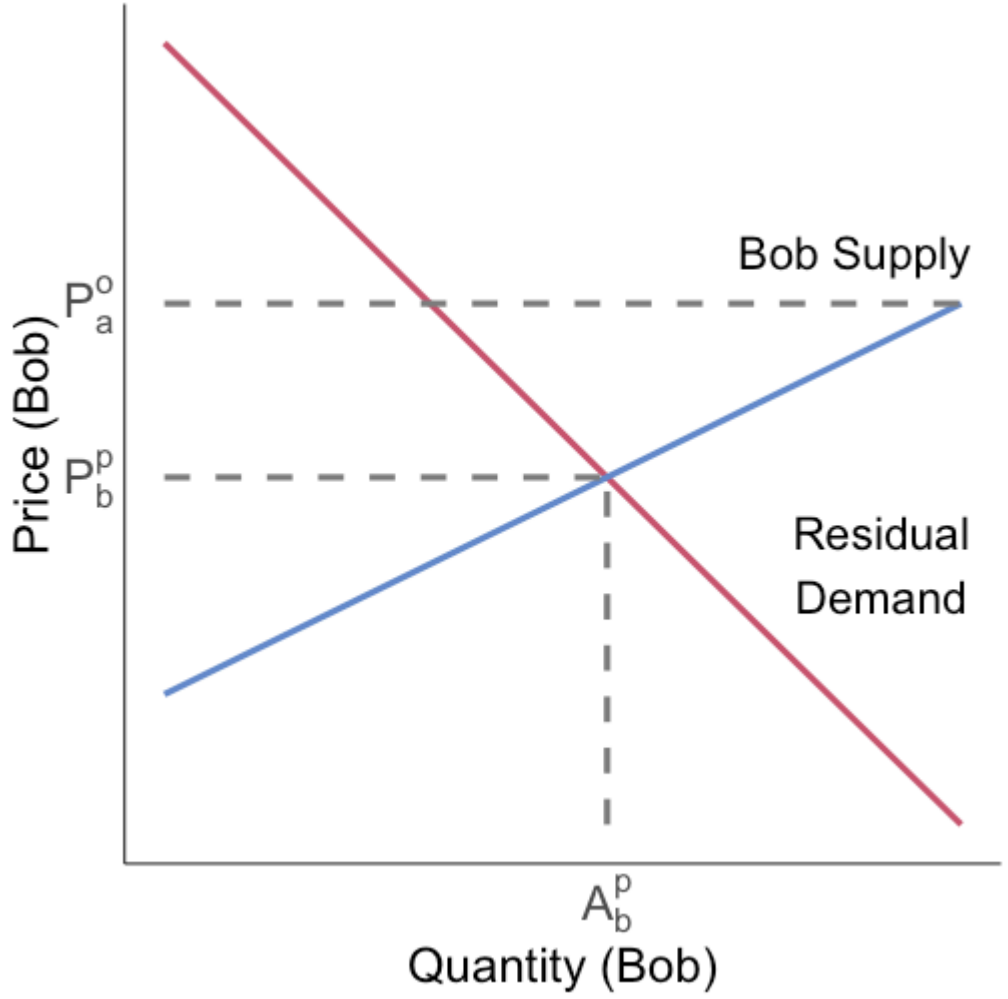

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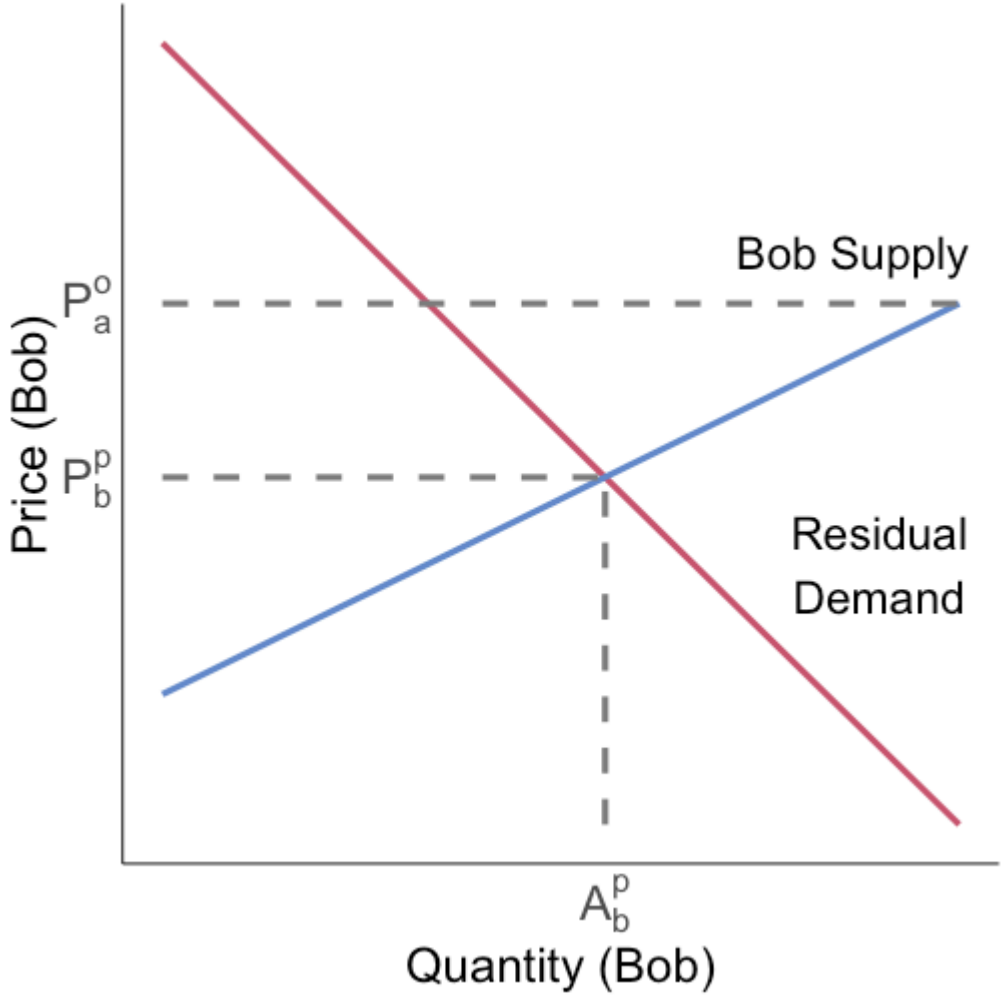
**But,** they can still buy from Bob!

People who were willing to pay less than $P^o$ increase Bob's residual demand



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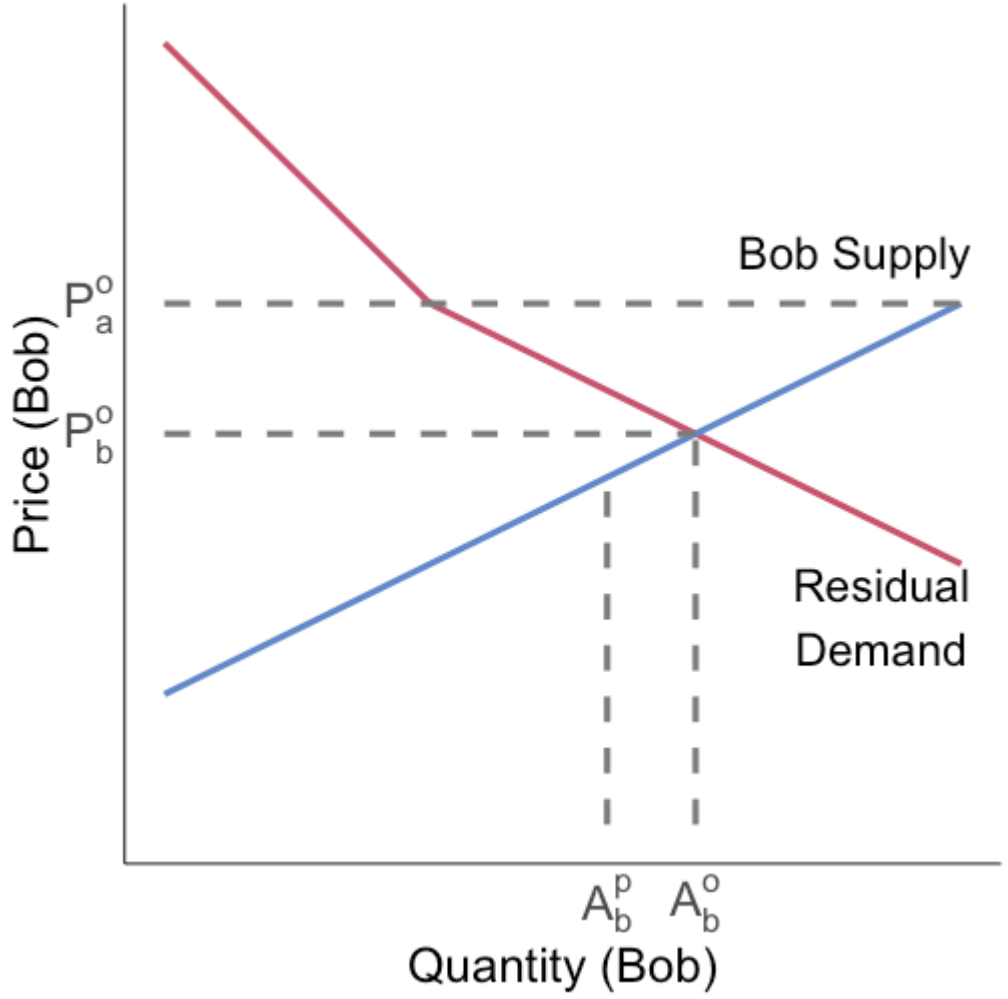
This shifts out Bob's demand at prices less than $P_a^o$

Bob's price rises to $P_b^o$

Bob then puts $A_b^o - A_b^p$ extra units of land into cotton production

$A_b^o - A_b^p$ units of carbon "leaked" from Ann to Bob



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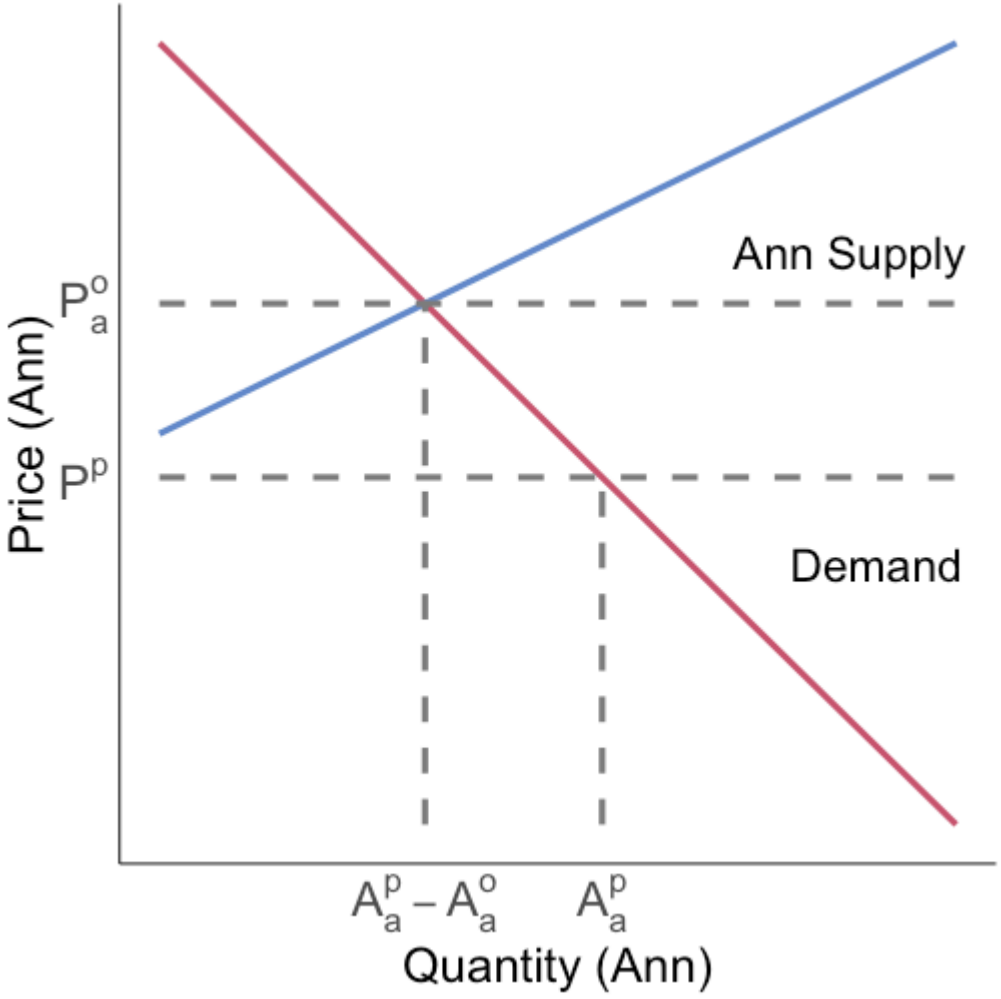


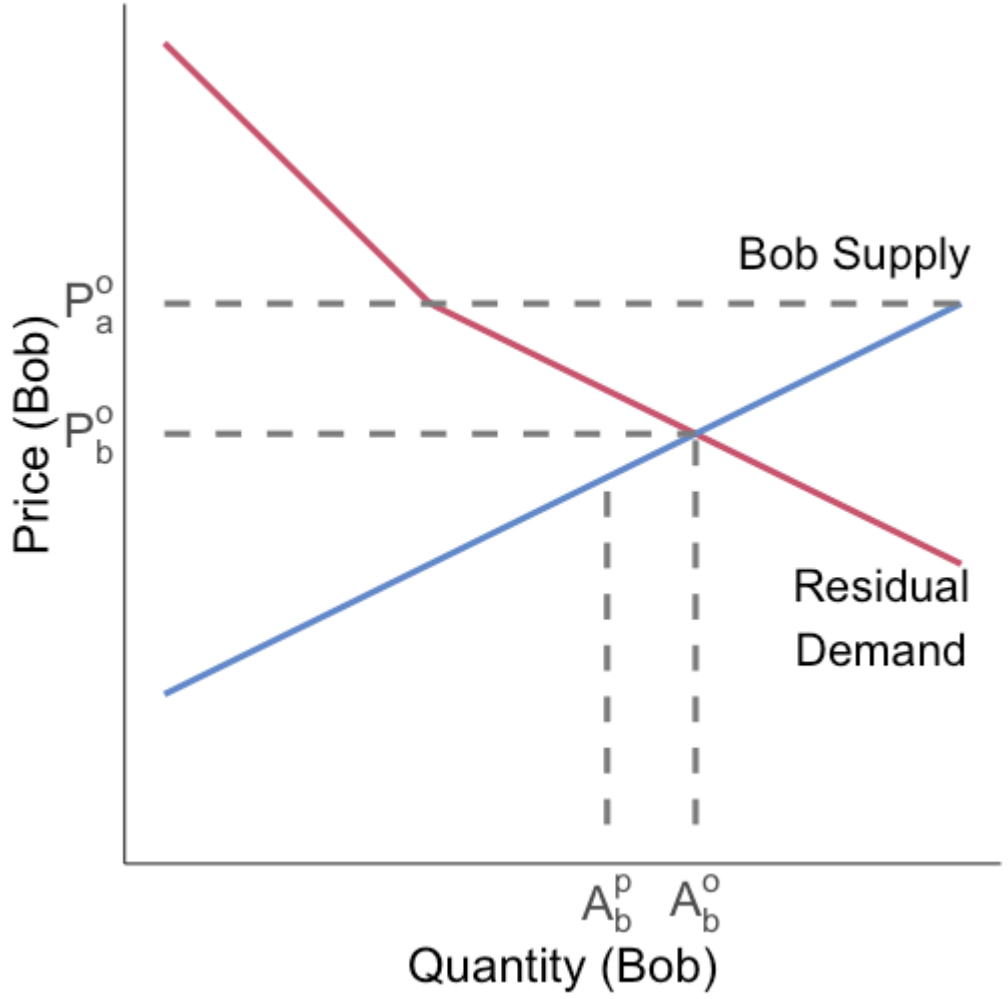

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Depends on elasticities of supply and demand

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Leakage matters when supply and demand are elastic, but not perfectly elastic

# How do offsets work in practice?

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CDM is projected to issue 12 billion offsets by 2030

- Equivalent to current emissions of US and Europe combined

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Let's think through potential issues with this approach to evaluating additionality

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- Is the average power plant the one being displaced by the new project today?
- Should we expect emissions to be the same in the future?

Let's see how well the CDM worked in India: Calel et al. (2022)

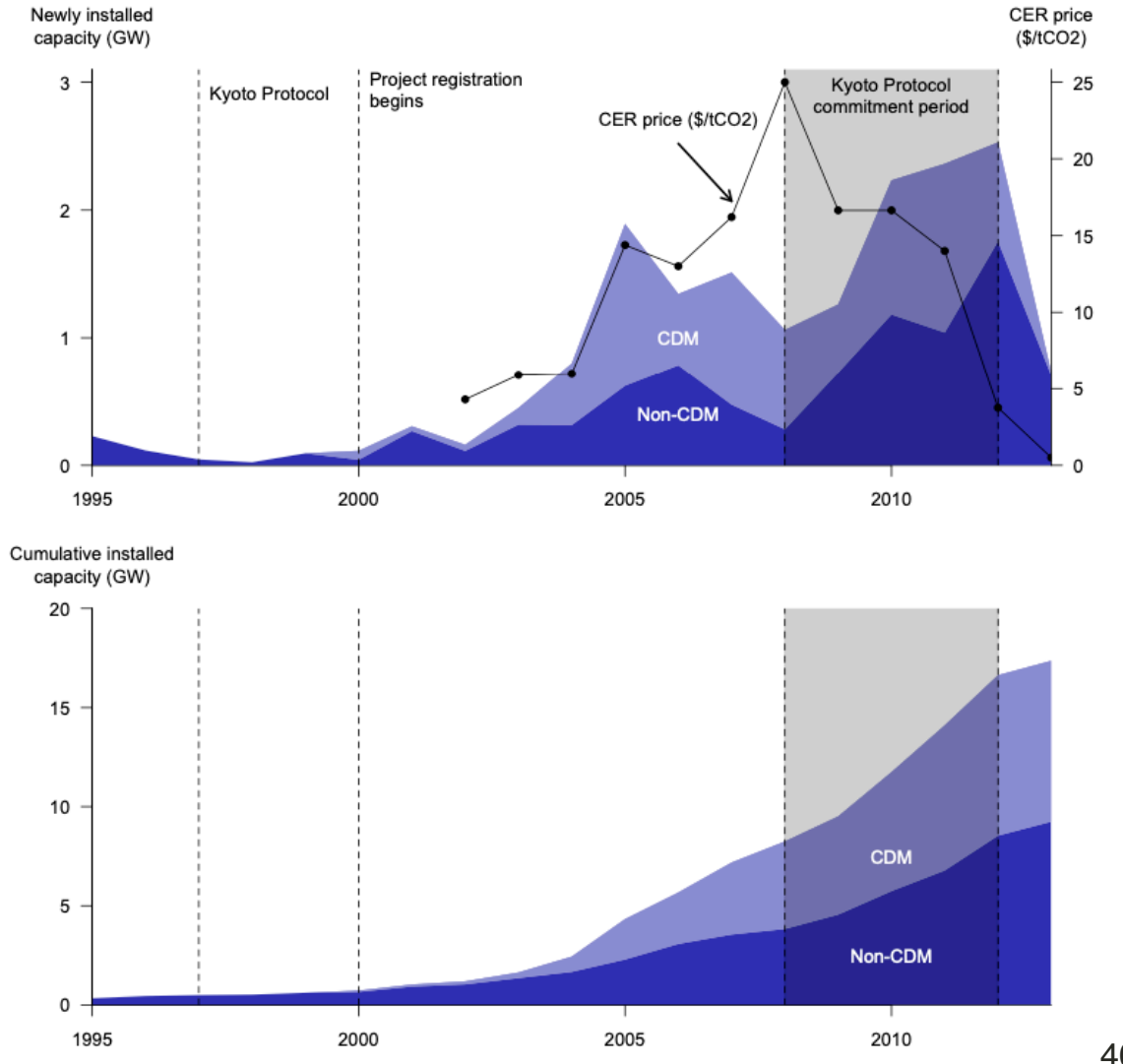
# Evaluating CDM additionality: wind projects

Half of projects generated CERs (~50 million total expected CERs)

20x'ed the size of India's wind capacity

Wind is an area where CDM was expected to work well

- Easy target: Chinese/Russian firms manufactured HFC-23 just to destroy it to get CERs



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These are projects that are obviously not additional based on observable data

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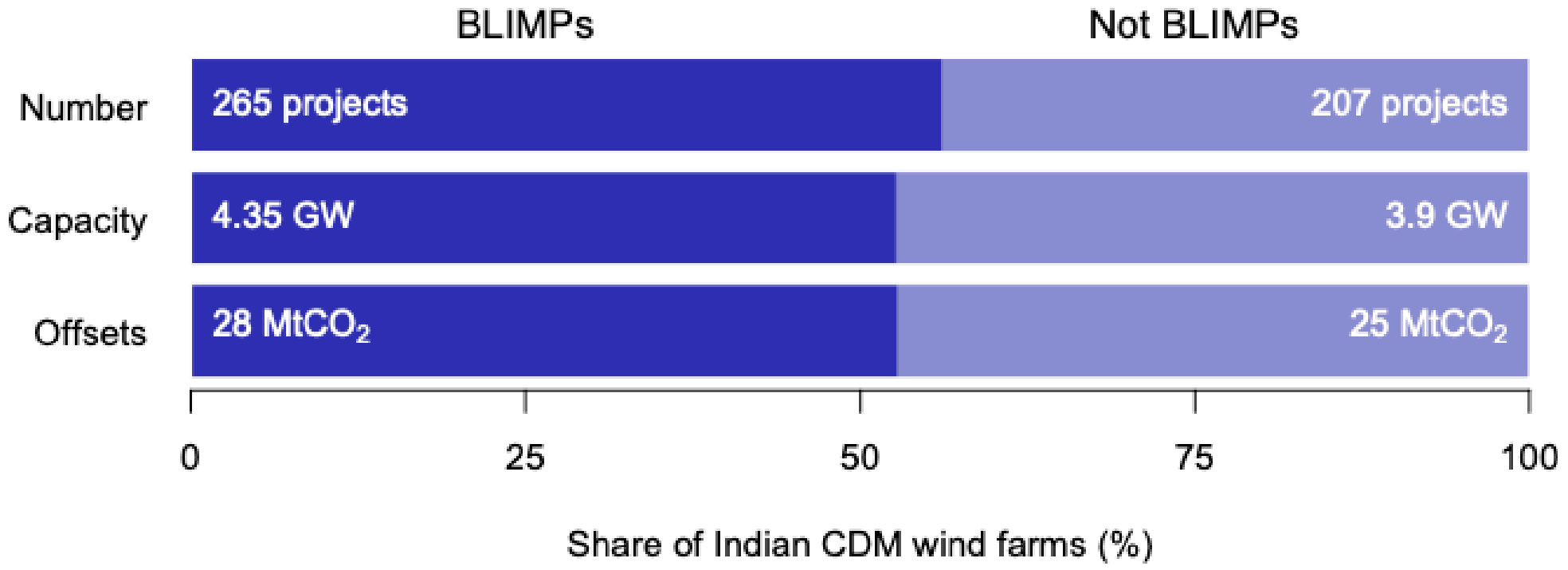
If $X$ is built without any CER revenues, then $Y$ must not be additional

What they do is go through each CDM project and try to find a clearly "better" project that was built without any CER payments

- "Better" is bigger, and windier, and closer to a connection point

# India is BLIMP-city (i.e. SUPER not additional)




Half of the projects, capacity, and offsets in India are BLIMPs

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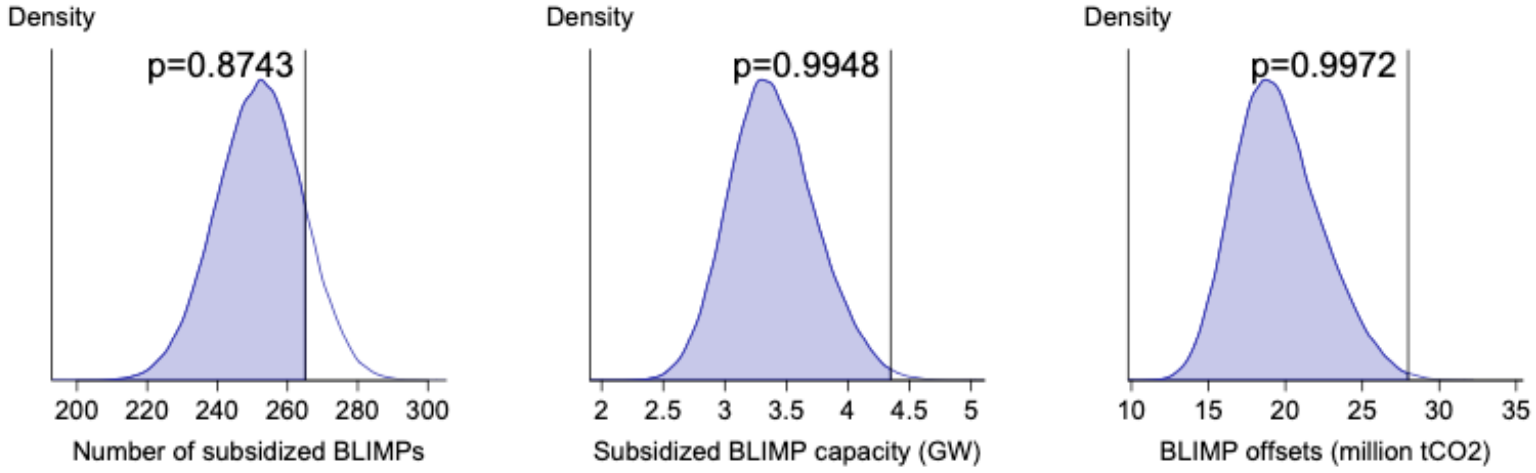
Half the emissions reductions were from projects that were more profitable than existing non-CDM projects: they were going to be built even without the CDM

Lets compare this against a lottery: randomly assigning CDM-approved status, holding fixed the total number of CDM projects

This is a conservative benchmark, isn't even trying to target additional projects

# How bad is it? Bad




87% of the time a lottery generates fewer BLIMP projects

99% of the time the lottery subsidizes less BLIMP capacity and CERs

# CDM takeaways

The current way we do CDM in India is not good

- 653 out of 666 proposed projects were approved for CDM

We need to be very thoughtful about how we allocate valuable funding if we want it to have a real impact